Code-Layout, Readability and Reusability

Date 05 October 2026

Team ID UNM-293

Project Name FitBuddy-AI fitness plan Generator using Gemini Models

Project ID 2026-2430

Name Kaviya sree.T

Maximum Marks 5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the FitBuddy codebase in terms of its structure, readability, and reusability. The implementation should use meaningful naming, modular design, clear comments, consistent formatting, and reusable components so that the Gemini-based fitness plan generation module can be maintained and extended.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the FitBuddy code.	Yes	Consistent formatting is followed.
2	Proper File Structure	Files and folders are logically organized by application responsibility.	Yes	UI, backend, AI integration and supporting files are separated.
3	Meaningful Variable Names	Variables clearly reflect user inputs, prompts, responses and fitness-plan data.	Yes	Descriptive names are used.
4	Function / Method Names	Functions are descriptively named according to their task.	Yes	Names indicate the purpose of each function.
5	Code Comments	Inline and block comments explain important logic and Gemini integration.	Yes	Comments are added where the logic needs clarification.
6	Modular Design	Code is split into reusable functions/modules for input handling, prompt creation and AI response processing.	Yes	Core AI generation logic is kept modular.
7	No Redundant Code	Duplicate or unused code is removed from the project.	Yes	Common operations are reused instead of duplicated.
8	Error Handling	Exceptions and API-related errors are handled gracefully.	Yes	Invalid inputs and Gemini service failures are handled with suitable messages.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	User Input Handler	Python / Web technology	Collecting and validating fitness goals, preferences and requirements.	High
2	Prompt Builder	Python	Creates structured prompts for different fitness-plan generation requests.	High
3	Gemini API Service	Python + Gemini API	Reusable service for sending prompts and receiving AI-generated responses.	High
4	Fitness Plan Formatter	Python	Formats Gemini output into a consistent, readable fitness-plan structure.	High
5	Error Handling Module	Python	Handles invalid inputs, API failures and unexpected responses.	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized structure separating input handling, AI integration, processing and output.
Readability	5	Meaningful names, consistent formatting and clear organization make the code easy to understand.
Reusability	5	Prompt, Gemini API, formatting and validation modules can be reused across different fitness-plan requests.
Documentation / Comments	4	Important logic and AI integration steps are documented with useful comments.
Overall Score	4.8 / 5	Strong overall code quality with modular and reusable AI fitness-plan generation components.

Assigned Work – Kaviya sree.T

Primary responsibility: Development and quality of the FitBuddy fitness-plan generation module. This includes writing readable and modular code, creating reusable prompt-building and Gemini API functions, formatting AI-generated fitness plans, adding error handling, testing the module, and maintaining documentation/comments.